Marking Scheme

Module Code: COMP I8064

Module Title: Data Structures

Internal Examiner(s): Dr Michelle Graham

Session: SAMPLE PAPER

Year: 24-25

Instructions:

Answer QUESTION ONE (40 marks) and any other TWO questions (30 marks each).

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You MUST attempt this question.

Question 1) Data Structure Fundamentals

(a) Data structures are a way of storing / organising information so that it can be used effectively and efficiently later.

(i) For each of the below data structures, give a real-world example where it would be a suitable choice. Provide a one sentence explanation for why this is the case.

- Binary Search Tree
- Queue
- List

(6 marks)

SAMPLE ANSWER:

Any options provided that make sense based on the explanation, assuming there is no obvious factor negating the explanation. For example, storing Users in a queue because you can get the Users in order of when they registered – yes that is what a queue would be used for, but it would not be a good way to store Users overall.

MARKS BREAKDOWN:

Example: 1 mark * 3

Explanation: 1 mark * 3

(ii) Give a real-world example of where each of the below data structures would be a poor choice. Provide a one sentence explanation for why this is the case.

- Stack
- Set
- HashMap

(6 marks)

SAMPLE ANSWER:

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Any options provided that make sense based on the explanation, assuming there is no obvious factor negating the explanation. For example, not storing Users in a hashmap because their username could change – yes the username changing would mean it's a bad key to use, but you could use a different attribute as the key instead (or even create one)

MARKS BREAKDOWN:

Example: 1 mark * 3

Explanation: 1 mark * 3

(b) There are many data structures to choose from when programming. Identify three criteria that should be considered when choosing a data structure for use in an application. Explain why these criteria are of such importance.

(6 marks)

SAMPLE ANSWER:

- Time complexity – speed concerns
- Space complexity – available memory concerns
- System requirements – is random access required? Is retrieval a priority? Does the order need to be preserved? Should duplicates be allowed? Etc

MARKS BREAKDOWN:

- Outlining criterion: 2 * 1.5 marks
- Explaining importance 2 * 1.5 marks

(c) Generics are a special form of data structure designed for flexibility and reuse.

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(i) **Define** a generic data structure and **explain** how it differs from a non-generic data structure. Provide an example to illustrate your answer.

(5 marks)

SAMPLE ANSWER:

Must mention the ability to hold any type of data vs a fixed type.

Must mention reusability.

MARKS BREAKDOWN:

Definition: 2 marks

Explanation: 2 marks

Example: 1 mark

(ii) Explain one benefit and one potential drawback to implementing a data structure as a generic version vs implementing using a strongly-typed approach.

(5 marks)

MARKS BREAKDOWN:

Benefit: 2.5 marks

Drawback: 2.5 marks

(d) Your team has been asked to write unit tests for a HashMap that manages collisions using collision chaining. The map holds String-User key-value pairs, with the key being the User's username (a unique and unchanging value). You have been assigned the following method:

- public int put(String s, User u)

Outline three test cases that should be included as unit tests for the above method. For each test case, explain its necessity, what assertions should be included and why they are needed.

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(12 marks)

SAMPLE ANSWER:

Potential test cases:

- Adding a null key
- Adding a key with a colliding hash value
- Updating an existing key
- Adding the first key-value pair to a slot
- Other valid test case options

MARKS BREAKDOWN:

- Outlining and explaining test case: 3 * 2 marks
- Listing and explaining need for assertions 3 * 2 marks

Choose any TWO questions from questions 2, 3 and 4

Question 2) Binary Search Trees

(a) What is a "Binary Search Tree" data structure? List and explain the fundamental operations for a *Binary Search Tree*.

(10 marks)

MARKS BREAKDOWN:

Binary Search Tree explanation (MUST sorted nature of data and requirement for data to be orderable): 2 marks

Operations (add, remove, contains, min & max operations, iterator): 4 marks

Explanations: 4 marks

(b) Write and explain the **pseudocode** for the *add* method in a Binary Search Tree. The Tree should contain String objects. Note: Your method should be done as pseudocode, not actual code. Pseudocode and explanation are worth equal marks.

(8 marks)

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MARKS BREAKDOWN:

Pseudocode: 4 marks

Explanation: 4 marks

(c) Write and explain the **pseudocode** for the *removeMin* method in a Binary Search Tree. The Tree should contain String objects, and the removed value should be returned. Note: Your method should be done as pseudocode, not actual code. Pseudocode and explanation are worth equal marks.

(12 marks)

MARKS BREAKDOWN:

Note regarding need for wrapper class to allow value to be returned. 2 marks

Pseudocode: 5 marks

Explanation: 5 marks

Question 3) Maps

(a) A Hash Map is a commonly used data structure that allows for fast data retrieval and storage. How is a Hash Map different from a standard array?

(4 marks)

MARKS BREAKDOWN:

General overview of Hash Map: 1 marks

Difference from array: 3 marks

(b) What is meant by the term *collision* in relation to a Hash Map? Explain the main causes of collisions in Hash Maps. What is the impact of *collisions* in a Hash Map without collision resolution?

(6 marks)

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MARKS BREAKDOWN:

Explanation: 2 marks

Causes: 2 marks

Impact: 2 marks

(c) The **load factor** is an important concept impacting Hash Map performance and efficiency.

(i) What is the load factor of a hash map? Explain how it is calculated and what it represents.

(6 marks)

SAMPLE ANSWER:

Load factor: How full the map is relative to the length of inner array

High load indicates chance of collisions is high

MARKING SCHEME:

Definition: 2 marks

Calculation: 2 marks

Representation: 2 marks

(ii) How does a higher load factor impact the **performance** of a Hash Map? Your answer should include a discussion of the lookup & insertion actions.

(8 marks)

SAMPLE ANSWER:

Higher load factor threshold increases chance of collisions as more of the map can be occupied. Higher chance of collisions means cost of search and insertion is higher, as there is more cause for iteration to process the collisions.

MARKING SCHEME:

Explanation: 2 marks

Lookup discussion: 3 marks

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Insertion discussion: 3 marks

(iii) What is **rehashing**, and why is it necessary in Hash Maps? Explain the link between load factor and rehashing.

(6 marks)

SAMPLE ANSWER:

Rehashing: Recomputing the position at which an element should be placed within a map. It is done when the map reaches capacity and needs to be expanded. Low load factor equals more frequent rehashing.

MARKING SCHEME:

Definition: 1 mark
Necessity: 3 marks
Link: 2 marks

Question 4) Analysis of Data Structures

(a) Your team has been assigned the task of building a ticketing system to store information on raised issues (stored as Ticket objects). You have been asked to research suitable data structures for the storage of Ticket information in memory (a database approach has been deemed unsuitable for this situation).

(i) Compare and contrast the suitability of the following structures as ticket storage for this system:

- Collision-resistant HashMap
- Tree-based Priority Queue
- Link-based Priority Queue

Your answer should include a discussion of time and space complexities. It should consider:

(i) Insertion.

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(ii) Deletion.

(iii) Retrieval.

(27 marks)

All must be placed in context of ticket storage.

All must mention big O.

Insertion: 3 marks * 3

Deletion: 3 marks * 3

Retrieval: 3 marks * 3

(iv) Recommend, with reference to your answer to 4a) which structure should be employed if the system is to be deployed in a low memory environment where Tickets are frequently required to be retrieved in order of creation date.

(3 marks)

MARKS BREAKDOWN:

Verdict: 1 mark

Justification: 2 marks

Scope for allowable answers is broad. Acceptance is based on justification provided.

Memory constraints **must** be mentioned.

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